

Determinants of Service Quality and Their Impact on Inpatient Satisfaction at Doloksanggul Hospital in 2025

Determinan Kualitas Pelayanan Dan Dampaknya Terhadap Kepuasan Pasien Rawat Inap di Rumah Sakit Doloksanggul Tahun 2025

Rumondang Pandiangan ^{a*}, Kesaktian Manurung ^a, Viero Irensus Girsang ^a, Frida Lina Traigan ^a,
Donal Nababan ^a

^a Master of Public Health Study Program, Faculty of Pharmacy and Health Sciences, Sari Mutiara Indonesia University, Medan, Indonesia.

*Corresponding Authors: rumondangpandiangan78@gmail.com

Abstract

Patient-reported service quality can identify priorities for improving inpatient care, but its association with satisfaction must be distinguished from a causal effect. This cross-sectional study examined the five *SERVQUAL* dimensions and inpatient satisfaction at Doloksanggul Regional General Hospital, North Sumatra, Indonesia. A purposive sample of 245 respondents completed a structured questionnaire. Descriptive statistics, Pearson's chi-square tests, and Cramér's V were used, and multiple logistic regression was attempted with all five dimensions entered simultaneously. Men accounted for 50.6% of respondents; intermediate education was the largest educational category (45.3%), and 46.1% had been hospitalized for 2–5 days. Overall, 126 respondents (51.4%) were satisfied. High service-quality ratings ranged from 50.6% for assurance to 53.1% for tangibles. All five dimensions were associated with satisfaction (all $p < 0.001$), with Cramér's V ranging from 0.837 for tangibles to 0.984 for assurance. Satisfaction was reported by 124/124 respondents with high assurance and 2/121 with low assurance. However, the logistic model did not converge after 20 iterations, and collinearity diagnostics indicated substantial overlap among predictors. Consequently, interpretable adjusted odds ratios and a defensible ranking of independent determinants could not be established. The findings describe strong unadjusted associations between perceived service quality and satisfaction. Assurance, responsiveness, and the substantial proportion of dissatisfied respondents warrant attention in service evaluation, but intervention priorities should be supported by verified measurement, operational indicators, and prospective assessment.

Keywords: Service quality; *SERVQUAL*; Inpatient satisfaction; Assurance; Responsiveness; Doloksanggul.

Abstrak

Mutu pelayanan yang dilaporkan pasien dapat membantu mengidentifikasi prioritas perbaikan layanan rawat inap, tetapi hubungannya dengan kepuasan perlu dibedakan dari pengaruh sebab-akibat. Penelitian potong lintang ini mengkaji lima dimensi *SERVQUAL* dan kepuasan pasien rawat inap di Rumah Sakit Umum Daerah Doloksanggul, Sumatera Utara, Indonesia. Sebanyak 245 responden dipilih secara purposif dan mengisi kuesioner terstruktur. Analisis mencakup statistik deskriptif, uji chi-square Pearson, dan Cramér's V; regresi logistik berganda juga dilakukan dengan memasukkan kelima dimensi secara simultan. Responden laki-laki berjumlah 50,6%; kategori pendidikan sedang merupakan kelompok pendidikan terbesar (45,3%), dan 46,1% responden menjalani perawatan selama 2–5 hari. Secara keseluruhan, 126 responden (51,4%) menyatakan puas. Proporsi penilaian mutu pelayanan berkategori tinggi berkisar antara 50,6% pada jaminan dan 53,1% pada bukti fisik. Kelima dimensi berhubungan dengan kepuasan pasien (seluruh $p < 0,001$), dengan Cramér's V berkisar antara 0,837 pada bukti fisik dan 0,984 pada jaminan. Kepuasan dilaporkan oleh 124/124 responden dengan penilaian jaminan tinggi dan 2/121 responden dengan penilaian jaminan rendah. Namun, model regresi tidak mencapai konvergensi setelah 20 iterasi, dan pemeriksaan kolinearitas menunjukkan tumpang tindih yang besar antarprediktor. Oleh karena itu, odds ratio tersesuaikan yang dapat ditafsirkan dan urutan determinan independen belum dapat ditetapkan. Temuan ini menggambarkan hubungan tanpa penyesuaian yang kuat antara persepsi mutu pelayanan dan kepuasan. Jaminan, daya tanggap, dan besarnya proporsi pasien tidak puas perlu mendapat perhatian dalam evaluasi layanan, tetapi penetapan prioritas intervensi memerlukan pengukuran yang terverifikasi, indikator operasional, dan penilaian prospektif.

Kata kunci: Mutu pelayanan; *SERVQUAL*; Kepuasan pasien rawat inap; Jaminan; Daya tanggap; Doloksanggul.



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Introduction

Patient satisfaction provides an important account of how healthcare is experienced, particularly during hospitalization, when patients encounter multiple professionals and service processes. The World Health Organization's quality framework emphasizes safe, effective, people-centered, timely, equitable, integrated, and efficient care [1,2]. In this framework, satisfaction is informative but does not, by itself, establish clinical effectiveness or safety. The distinction is relevant to hospital quality management [3,4]. Contemporary clinical medicine likewise considers quality, safety, and value together, as reflected in the organization of Goldman-Cecil Medicine [5].

Satisfaction is shaped by patients' expectations and their appraisal of the care received. Service-marketing perspectives therefore distinguish the evaluation of a particular encounter from broader perceptions of service quality [6–8]. For inpatient services, these evaluations can encompass the physical environment, consistency of care, responsiveness to requests, confidence in staff, and individualized attention. A systematic review by Ferreira et al. documented substantial diversity in both the determinants of satisfaction and the instruments used to measure it [4]. Comparisons across hospitals consequently require attention to the population, outcome definition, and measurement approach rather than to a satisfaction percentage alone.

The *SERVQUAL* framework organizes perceived service quality into tangibles, reliability, responsiveness, assurance, and empathy [9,10]. Tangibles concern the visible care environment; reliability concerns dependable service delivery; responsiveness addresses timely assistance; assurance encompasses perceived competence, courtesy, and trust; and empathy concerns individualized attention. These related constructs provide a useful framework for organizing patient feedback. Nevertheless, a study based on categorized perceptions is not equivalent to an analysis of the difference between paired expectation and perception scores. Maintaining this distinction is important when interpreting *SERVQUAL*-informed hospital research [11,12].

Recent Indonesian studies illustrate why the relative importance of these dimensions should not be assumed to be uniform. Rahmatia et al. reported associations between all five dimensions and inpatient satisfaction at Batara Siang Hospital, with empathy showing the largest regression coefficient [12]. In contrast, Widyaputri et al. found associations for tangibles, responsiveness, and reliability among non-national-health-insured inpatients, but not for assurance or empathy [13]. These differences support context-specific evaluation; they do not establish a universal hierarchy of service-quality dimensions.

Doloksanggul Regional General Hospital (Rumah Sakit Umum Daerah Doloksanggul; RSUD Doloksanggul) is a government-owned Type C referral hospital in Humbang Hasundutan Regency, North Sumatra. Its inpatient services operate alongside a patient-liaison initiative known as Pramugari Jaminan Kesehatan Nasional (JKN), which assists patients and families with service information and complaints. This setting provides an opportunity to examine how patients appraise interpersonal and organizational aspects of care within a district referral service. The present study did not measure individual exposure to the liaison initiative and was not designed to evaluate its effectiveness.

The study's contribution is a local assessment of the distribution of perceived service quality, the strength of its unadjusted associations with satisfaction, and the feasibility of estimating independent associations. The practical question is not simply whether statistical associations exist, but whether the available measurements can distinguish priorities sufficiently to guide improvement. Accordingly, the study examined the five *SERVQUAL* dimensions in relation to inpatient satisfaction at RSUD Doloksanggul and assessed the limitations of their joint statistical interpretation.

Research questions and hypotheses

The research questions were whether inpatient satisfaction differed between high and low ratings of each service-quality dimension and whether the dimensions retained independent associations when considered jointly. The corresponding hypotheses were that satisfaction was associated with tangibles (H1), reliability (H2), responsiveness (H3), assurance (H4), and empathy (H5). A joint logistic model was intended to investigate independent associations, without assuming that the dimension with the largest unadjusted association would necessarily be the dominant adjusted determinant.

Methods

Study design and setting

This analytical cross-sectional study was conducted in the inpatient services of RSUD Doloksanggul, Humbang Hasundutan Regency, North Sumatra, Indonesia. Service-quality perceptions and satisfaction were assessed within the same survey. The reported study period extended from October 2025 to February 2026. Because the design did not establish temporal ordering between the exposure and outcome assessments, the analyses addressed associations rather than causal effects.

Participants and sample size

The target population comprised patients receiving inpatient care at the hospital. Administrative data for January–June 2025 recorded 3,800 inpatient admissions, corresponding to approximately 633 admissions per month. The planned sample was approximately 245 respondents, based on $n = N/[1 + N(e^2)]$, with $N = 633$ and $e = 0.05$. A total of 245 respondents were included through purposive sampling. This calculation was used for sample planning; the non-probability recruitment approach does not support a design-based claim that the estimates have a 5% sampling margin of error.

Measures and questionnaire

Primary data were collected using a structured questionnaire administered through guided interviews. The questionnaire addressed respondent characteristics, the five *SERVQUAL* dimensions, and satisfaction with inpatient care. The perception form contained four items each for tangibles, reliability, responsiveness, and assurance; five items for empathy; and one overall-satisfaction item. It used five ordered response options from strongly disagree to strongly agree. The original Indonesian questionnaire, including its accompanying expectation section, is provided in Supplementary Material S1.

Tangibles addressed cleanliness, facilities, comfort, and staff appearance. Reliability concerned consistent and accurate service delivery; responsiveness concerned timely assistance and responses to complaints. Assurance addressed confidence in staff competence, respectful treatment, and understandable information. Empathy concerned individualized attention and consideration of patients' needs. The analytical dataset classified each service-quality dimension as high or low and satisfaction as satisfied or dissatisfied. The analysis used these recorded categories rather than a perception-minus-expectation gap score.

Instrument properties were considered separately from the analysis of the categorical outcomes. The archived analytical file did not contain individual questionnaire-item responses; consequently, internal consistency, factor structure, and discriminant validity could not be evaluated in the present analysis. The findings are therefore interpreted as associations between the recorded service-quality classifications and satisfaction, rather than as validation of a locally calibrated *SERVQUAL* scale.

Statistical analysis

Categorical variables were summarized as frequencies and percentages. Associations between each service-quality dimension and satisfaction were assessed using Pearson's chi-square test with one degree of freedom. Expected cell frequencies were examined to assess the adequacy of the asymptotic test. Cramér's V was calculated as $\sqrt{(\chi^2/n)}$ for each 2×2 table to describe the strength of association. Percentages in the cross-tabulations were calculated within each service-quality category. All tests were two-sided, with $p < 0.05$ indicating statistical significance; values displayed as 0.000 by the statistical software are reported as $p < 0.001$.

Multiple binary logistic regression was attempted with all five service-quality dimensions entered simultaneously using the Enter method. The dependent variable was encoded as 0 for satisfied and 1 for dissatisfied. Model evaluation included inspection of convergence, coefficient stability, separation, and the

available goodness-of-fit output. Post hoc reproducibility checks of the archived 245-record categorical dataset included code-range checks, reconciliation of frequencies with the statistical output, and calculation of variance inflation factors (VIFs) among the five predictors. Adjusted odds ratios were not interpreted when the estimation procedure failed to converge. Separation can produce divergent maximum-likelihood estimates and requires a different inferential approach from routine interpretation of the last iteration [14].

All 245 records had non-missing values for the analyzed categorical variables, so no imputation was required. The available variables contained only the expected category codes. No observation was removed solely because it shared a categorical response pattern with another participant. Between-group homogeneity testing was not performed because the study described a single purposively recruited cohort rather than predefined comparison arms.

Results and Discussion

Participant characteristics

The analysis included 245 respondents. The distribution by sex was nearly balanced, with 124 men (50.6%) and 121 women (49.4%). Intermediate education was the largest educational category (111; 45.3%), followed by low education (76; 31.0%) and high education (58; 23.7%). More than half of the respondents were employed (139; 56.7%). The most frequent length of stay was 2–5 days (113; 46.1%); 80 respondents (32.7%) had stayed for less than 2 days and 52 (21.2%) for more than 5 days (Table 1).

Table 1. Characteristics of the study participants (n = 245).

Characteristic	Category	n	%
Sex	Male	124	50.6
	Female	121	49.4
Education	Low	76	31.0
	Intermediate	111	45.3
	High	58	23.7
Employment	Employed	139	56.7
	Not employed	106	43.3
Length of stay	<2 days	80	32.7
	2–5 days	113	46.1
	>5 days	52	21.2

Service-quality ratings and overall satisfaction

High ratings were reported by 130 respondents (53.1%) for tangibles, 126 (51.4%) for reliability, 128 (52.2%) for responsiveness, 124 (50.6%) for assurance, and 128 (52.2%) for empathy (Figure 1). Thus, ratings were divided almost equally between the high and low categories across all five dimensions. Overall, 126 respondents (51.4%) were satisfied and 119 (48.6%) were dissatisfied. These distributions do not indicate uniformly favorable evaluations, even though slightly more than half of the respondents were classified as satisfied.

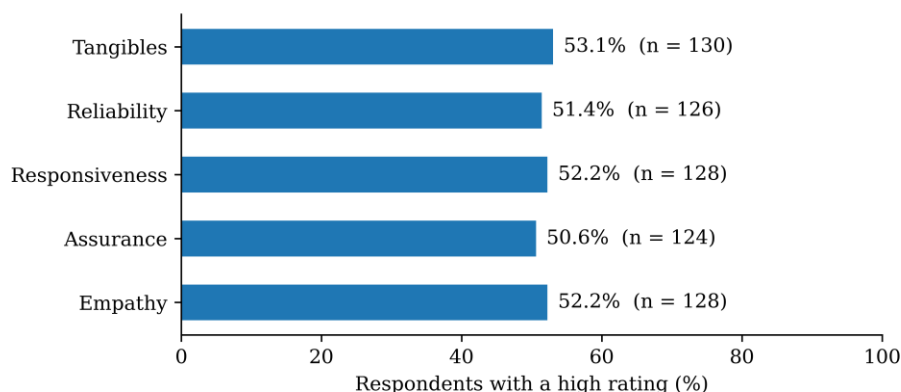


Figure 1. Proportion of respondents with high ratings for each service-quality dimension (n = 245). Counts are shown beside the percentages. Source: study data.

Bivariate associations

All five service-quality dimensions were associated with satisfaction (all $p < 0.001$; Table 2). Cramér's V ranged from 0.837 for tangibles to 0.984 for assurance, indicating very strong unadjusted associations in the recorded categorical data. Satisfaction was reported by 90.8% of respondents with high tangibles ratings, compared with 7.0% of those with low ratings. The corresponding proportions were 96.8% versus 3.4% for reliability, 96.9% versus 1.7% for responsiveness, and 95.3% versus 3.4% for empathy. For assurance, all 124 respondents in the high category were satisfied, whereas only 2 of 121 respondents (1.7%) in the low category were satisfied.

Table 2. Service-quality dimensions and inpatient satisfaction (n = 245).

Dimension	Rating	Satisfied, n (%)	Dissatisfied, n (%)	Total, n	χ^2	V	p
Tangibles	High	118 (90.8)	12 (9.2)	130	171.597	0.837	<0.001
	Low	8 (7.0)	107 (93.0)	115			
Reliability	High	122 (96.8)	4 (3.2)	126	214.020	0.935	<0.001
	Low	4 (3.4)	115 (96.6)	119			
Responsiveness	High	124 (96.9)	4 (3.1)	128	221.618	0.951	<0.001
	Low	2 (1.7)	115 (98.3)	117			
Assurance	High	124 (100.0)	0 (0.0)	124	237.126	0.984	<0.001
	Low	2 (1.7)	119 (98.3)	121			
Empathy	High	122 (95.3)	6 (4.7)	128	206.641	0.918	<0.001
	Low	4 (3.4)	113 (96.6)	117			

Percentages are calculated within each high/low rating category. χ^2 = Pearson's chi-square statistic (df = 1); V = Cramér's V. Statistical values on the high-category row refer to the complete 2×2 table.

No expected cell frequency was below five; the minimum expected frequencies across the five tables ranged from 55.86 to 58.77. The observed zero cell for high assurance and dissatisfaction therefore did not invalidate the expected-frequency criterion for Pearson's test. It was, however, important for logistic estimation because it indicated separation in the relationship between assurance and the outcome. Figure 2 displays satisfaction within each rating category, using the category-specific denominator rather than the total sample as the denominator.

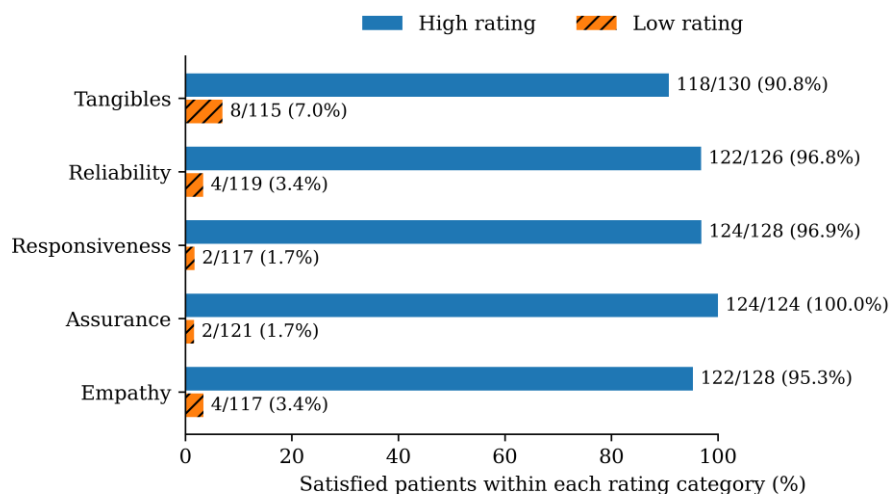


Figure 2. Satisfaction within the high- and low-rating categories of each service-quality dimension. Labels show satisfied respondents/category total and the corresponding within-category percentage. Comparisons are descriptive and unadjusted. Source: study data.

Joint-model estimation and diagnostic findings

The five-predictor logistic model did not converge within the specified limit of 20 iterations. The assurance coefficient increased markedly across iterations, while the other coefficients and their standard errors also became unstable. The final-iteration standard errors ranged from 8,546.158 to 29,424.765. These results did not support finite, interpretable adjusted odds ratios or confidence intervals. A final reduced model with reliable independent estimates was therefore not established.

Post hoc collinearity diagnostics also showed substantial overlap among predictors (Table 3). VIFs were 3.85 for tangibles, 10.78 for reliability, 15.87 for responsiveness, 35.40 for assurance, and 8.11 for empathy. In the categorical analytical file, the assurance and sex variables had identical category assignments across all 245 records. Consequently, assurance could not be distinguished from recorded sex in a model that included both variables. This correspondence further limits interpretation of assurance as an independently identified service-quality determinant.

Table 3. Collinearity diagnostics for the five service-quality predictors.

Predictor	VIF
Tangibles	3.85
Reliability	10.78
Responsiveness	15.87
Assurance	35.40
Empathy	8.11

VIF = variance inflation factor, calculated post hoc among the five categorical service-quality predictors with an intercept. The original logistic model did not converge; these diagnostics do not provide adjusted effect estimates.

The available Hosmer–Lemeshow statistic was $\chi^2 = 0.000$ with 2 degrees of freedom ($p = 1.000$), and the apparent classification accuracy was 99.2%. Neither statistic resolves failed convergence or establishes a valid predictive model. Accordingly, these values were not used to claim good model fit, predictive utility, or an independent ranking of the five dimensions. The supported inferential findings are the bivariate associations, interpreted alongside their measurement and coding limitations [14].

Interpretation in relation to previous evidence

The nearly even division between satisfied and dissatisfied respondents indicates that service evaluation should address both favorable experiences and unresolved concerns. Patient-satisfaction surveys are most useful when linked to specific, actionable service processes rather than treated as a single performance label [15,16]. Here, the small differences in the prevalence of high ratings across dimensions contrast with the extremely large differences in satisfaction within their categories. That pattern is consistent with strong coupling between the recorded classifications, but it cannot establish that changing one service dimension would produce a corresponding improvement in satisfaction.

The physical environment and dependable delivery of care remain relevant to patients' experiences, despite the inability to estimate their independent contributions in this dataset [13,17,18]. Facilities, medication schedules, communication about procedures, and consistency between promised and delivered services are distinct operational domains. For this study, however, the tangibles and reliability categories were patient-reported assessments rather than objective audits of cleanliness, equipment performance, or clinical accuracy. The observed associations should therefore not be interpreted as evidence that technical care was either safe or unsafe.

Assurance and responsiveness deserve particular attention because their high-rating categories coincided closely with satisfaction. Confidence in staff, understandable explanations, respectful conduct, and timely assistance are plausible areas for service improvement [12,19,20]. Nevertheless, assurance's largest Cramér's V describes an unadjusted relationship, not statistical proof that it is the most important independent determinant. The zero outcome cell, substantial collinearity, and identical assurance–sex coding prevent that stronger interpretation. Likewise, the non-converged model provides no basis for concluding that responsiveness has a specific adjusted effect size.

The comparison with Indonesian inpatient evidence is informative precisely because the findings are not uniform. Rahmatia et al. identified empathy as the dimension with the largest coefficient at Batara Siang Hospital [12], whereas Widyaputri et al. reported no significant assurance or empathy association in their sample of 97 non-national-health-insured inpatients [13]. Different sampling frames, outcome definitions, and statistical models limit direct comparisons of effect magnitudes. Neither study justifies transferring a numerical odds ratio or a fixed priority ranking to RSUD Doloksanggul.

Empathy should not be dismissed because an unstable joint model cannot separate its contribution from other perceptions [21]. Nembhard et al.'s systematic review supports attention to empathy in healthcare and identifies an extensive, but heterogeneous, research base on its measurement and improvement [22]. In the present study, 95.3% of respondents with high empathy ratings were satisfied, compared with 3.4% in the low

category. This association supports examining interpersonal care alongside other service processes; it does not establish mediation, moderation, or a causal pathway through assurance or responsiveness.

Other studies extend the discussion beyond satisfaction without making the outcomes interchangeable. Aryani et al. examined service quality and revisit intention in Mataram [23]; Panda et al. considered service quality, satisfaction, and loyalty in hospitals in Telangana [24]; and Hasan et al. examined outpatient services in Palopo [25]. These studies provide related service-management perspectives, but revisit intention, loyalty, outpatient experience, and inpatient satisfaction should not be treated as the same outcome. The current survey did not assess subsequent attendance, treatment adherence, or clinical outcomes.

Length of stay and respondent characteristics also require cautious interpretation. Cai et al., in a study of 2,587 inpatients in China, found that relationships between length of stay and satisfaction differed by service-quality dimension [26]. In the present dataset, length of stay describes exposure to inpatient services but provides no direct information on disease severity or the appropriateness of admission duration. Similarly, educational categories do not establish health-literacy levels. Sebera et al.'s district-hospital study in Rwanda highlighted communication and service delays while using a different population and satisfaction classification [27]. Such findings favor assessment of specific processes and case mix rather than unsupported generalizations from demographic categories.

Policy and service-improvement implications

For RSUD Doloksanggul, the most defensible managerial response is to connect patient feedback to observable processes and evaluate whether targeted changes improve experience. This is consistent with a broader service-administration perspective and a quality framework that considers patient-centeredness alongside safety and effectiveness [2,28]. The Pramugari JKN function could provide a channel for recording the type of request or complaint, the unit responsible, the response time, and whether the issue was resolved. These are proposed applications of the findings, not outcomes demonstrated by this study.

An operational improvement cycle could begin with a four-week baseline of call-response times, unresolved complaints, and patient understanding of the next steps in care. Ward leaders could then select locally feasible response targets, review performance monthly, and repeat the same independently administered satisfaction measure after three months. For assurance, the emphasis would be on understandable explanations and confirmation of patient comprehension, not on using favorable ratings as proof of professional competence. For responsiveness, the emphasis would be on documented response and resolution times. Facility maintenance, reliable medication-related communication, and individualized attention should remain part of the same improvement program rather than being displaced by a single "dominant" dimension.

Study limitations and future research

The cross-sectional design prevents causal inference, and purposive sampling limits generalizability beyond the participating patients. The number approached, reasons for non-participation, and detailed recruitment procedures were not available for quantifying selection bias. Perceptions of quality and satisfaction were collected in the same questionnaire, creating a risk of common-method variance and socially desirable responses. Because item-level data were unavailable, the extent of overlap between constructs could not be examined through factor analysis, and the effect of alternative scoring thresholds could not be assessed.

The strong associations are further constrained by sparse discordant cells, failed logistic convergence, and substantial predictor collinearity. The exact correspondence between assurance and recorded sex also prevents separation of those characteristics in the available dataset. Clinical condition, age, ward, insurance status, and prior expectations were not included in an interpretable adjusted analysis. These limitations preclude conclusions about independent determinants and limit the use of the estimated bivariate associations for allocating resources between service dimensions.

Future studies should establish a reproducible scoring protocol, verify the suitability of item wording for the full inpatient population, and assess the distinction between service-quality constructs and overall satisfaction. Prospective measurement at standardized points during admission and after discharge, together with operational service indicators, would help clarify temporal relationships. Comparative studies across district hospitals could assess transferability, while qualitative interviews could identify what patients mean by assurance and responsiveness. Following verification of the source records, separation-aware methods such as penalized logistic regression may be considered for an appropriately specified model [14].

Conclusions

In this sample of 245 inpatients at RSUD Doloksanggul, 51.4% were satisfied with care, and all five recorded service-quality dimensions showed strong unadjusted associations with satisfaction. Assurance had the largest bivariate association, followed by responsiveness, but the non-converged logistic model did not establish independent determinants or valid adjusted effect estimates. Service-improvement planning should therefore combine patient feedback with verified measurement and operational indicators. Clear communication, timely assistance, reliable care processes, appropriate facilities, and individualized attention remain relevant areas for evaluation, with their effects to be assessed prospectively.

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Conflict of Interest Statement

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